

Inverse-free Gram block dilations for every $\mathcal{C}_A$ operator

Candidate full answer to Question 5.9 of arXiv:2302.13873

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Status. Candidate full answer, likely valid, pending independent expert review. The construction covers singular Toeplitz Gram matrices and makes no closed-range or invertibility assumption.

1 Source question

Bhat, Ghatak, and Pamula, *Operator moment dilations as block operators*, arXiv:2302.13873, define $T \in \mathcal{B}(\mathcal{H})$ to be in $\mathcal{C}_A$, for a positive invertible $A \in \mathcal{B}(\mathcal{H})$, when the sequence

$$A^{-1/2}T^n A^{-1/2} \quad (n \geq 1)$$

admits a unitary dilation. At the end of Section 5 they ask [1, Question 5.9]:

How to write block decompositions for isometric and unitary dilations of general operators of the $\mathcal{C}_A$ class?

Question 5.9. *How to write block decompositions for isometric and unitary dilations of general operators of the $\mathcal{C}_A$ class?*

The source gives recurrence formulae for general moment sequences only when several defect blocks are boundedly invertible. The point below is that a finite Gram-space model replaces every such inverse by a quotient and an orthogonal complement.

2 Proof intuition

The moments already specify the inner products that the formal vectors $V^j h$ must have. For each n , put the first $n + 1$ formal copies of $\mathcal{H}$ into the Hilbert space obtained from their Toeplitz Gram matrix. Appending a zero and prepending a zero are isometries between consecutive Gram spaces; the latter maps are compatible with the former. Their inductive-limit shift is therefore the required minimal isometry.

Orthogonalizing the nested finite Gram spaces gives shells $\mathcal{D}_n$. Since the shift sends the first $n + 1$ moments into the first $n + 2$, its matrix on these shells is upper Hessenberg, and every block in column n is computed inside the finite Gram space of order $n + 1$. This is the desired inverse-free block algorithm. Finally, adjoining copies of the wandering defect $\mathcal{K}_+ \ominus V\mathcal{K}_+$ produces a negative identity-shift tail; one coupling column and the Hessenberg positive corner give the unitary block matrix.

3 The block construction

We use inner products linear in the second variable. Put

$$\zeta(0) = I, \quad \zeta(n) = A^{-1/2} T^n A^{-1/2} \quad (n > 0), \quad \zeta(-n) = \zeta(n)^*.$$

For $n \geq 0$ define the Toeplitz Gram operator

$$G_n = [\zeta(j-i)]_{i,j=0}^n \in \mathcal{B}(\mathcal{H}^{n+1}).$$

If $T \in \mathcal{C}_A$, then $G_n \geq 0$: indeed, for any unitary dilation U_0 of the moments and $x_0, \dots, x_n \in \mathcal{H}$,

$$\sum_{i,j=0}^n \langle x_i, \zeta(j-i)x_j \rangle = \left\| \sum_{j=0}^n U_0^j x_j \right\|^2.$$

Let E_n be the completion of $\mathcal{H}^{n+1}$ modulo the radical of the Gram form

$$\langle x, y \rangle_n = \sum_{i,j=0}^n \langle x_i, \zeta(j-i)y_j \rangle. \quad (1)$$

Equivalently, E_n is canonically $\overline{\text{ran } G_n^{1/2}}$ with the ambient Hilbert-space norm. The quotient-completion formulation is important: it is valid even when G_n is singular and $\text{ran } G_n^{1/2}$ is not closed.

On representatives define

$$J_n(x_0, \dots, x_n) = (x_0, \dots, x_n, 0), \quad S_n(x_0, \dots, x_n) = (0, x_0, \dots, x_n).$$

Both descend to isometries $E_n \rightarrow E_{n+1}$. For J_n this follows from the upper-left principal Gram block, and for S_n from the lower-right principal block and Toeplitz invariance. Moreover

$$S_{n+1}J_n = J_{n+1}S_n. \quad (2)$$

Let $\mathcal{K}_+ = \varinjlim (E_n, J_n)$, with canonical isometries $\kappa_n : E_n \rightarrow \mathcal{K}_+$. By (2),

$$V\kappa_n x = \kappa_{n+1}S_n x \quad (3)$$

defines an isometry on $\mathcal{K}_+$. Write $e_j h$ for the vector represented by h in coordinate j . Then

$$e_j h = V^j e_0 h, \quad e_i^* e_j = \zeta(j-i), \quad e_0^* V^n e_0 = \zeta(n). \quad (4)$$

In particular, e_0 embeds $\mathcal{H}$ isometrically and V is an isometric dilation of the $\mathcal{C}_A$ moment sequence. Set

$$\mathcal{K}_n = \kappa_n E_n = \overline{\text{span}}\{e_0 \mathcal{H}, \dots, e_n \mathcal{H}\}, \quad \mathcal{D}_0 = \mathcal{K}_0, \quad \mathcal{D}_n = \mathcal{K}_n \ominus \mathcal{K}_{n-1} \quad (n \geq 1).$$

Thus

$$\mathcal{K}_+ = \bigoplus_{n \geq 0} \mathcal{D}_n. \quad (5)$$

Let Q_i be the projection onto $\mathcal{D}_i$ and put

$$V_{ij} = Q_i V|_{\mathcal{D}_j} : \mathcal{D}_j \longrightarrow \mathcal{D}_i. \quad (6)$$

Because $V\mathcal{K}_j \subseteq \mathcal{K}_{j+1}$,

$$V_{ij} = 0 \quad (i > j + 1). \quad (7)$$

Hence, relative to (5),

$$V = \begin{bmatrix} V_{00} & V_{01} & V_{02} & V_{03} & \cdots \\ V_{10} & V_{11} & V_{12} & V_{13} & \cdots \\ 0 & V_{21} & V_{22} & V_{23} & \cdots \\ 0 & 0 & V_{32} & V_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots & \ddots \end{bmatrix}. \quad (8)$$

This is also a finite algorithm for the blocks. To compute column j , construct E_{j+1} from G_{j+1} , regard the earlier E_i as the nested closed ranges of the J_i , take their successive orthogonal differences $\mathcal{D}_i$, and compress S_j as in (6). Thus column j uses only G_{j+1} , orthogonal complements, and positive square roots. No inverse, pseudoinverse, or closed-range hypothesis occurs.

The first column recovers the familiar defect block. If $X = \zeta(1) = A^{-1/2}TA^{-1/2}$ and $D_X = (I - X^*X)^{1/2}$, then the map

$$e_1h - e_0Xh \mapsto D_Xh$$

extends to a unitary $\mathcal{D}_1 \simeq \overline{\text{ran } D_X}$. In these coordinates

$$V_{00} = X, \quad V_{10} = D_X. \quad (9)$$

The next moment gives an inverse-free characterization of the first off-diagonal row block:

$$V_{01}D_X = \zeta(2) - X^2. \quad (10)$$

Existence and boundedness follow from the Gram construction, and uniqueness holds because $\text{ran } D_X$ is dense in $\overline{\text{ran } D_X}$. The later blocks are obtained by exactly the same finite Gram orthogonalization.

4 Full theorem and proof

Theorem 1 (General $\mathcal{C}_A$ block dilations). *Let $A > 0$ be invertible and let $T \in \mathcal{C}_A$. The construction above has the following properties.*

1. The operator V of (3) is a minimal isometric dilation of the moment sequence

$$\{A^{-1/2}T^nA^{-1/2} : n \geq 1\}.$$

Relative to the canonical Gram shells $\mathcal{D}_n$, its blocks are given by (6) and have the upper-Hessenberg form (8). Every block in column j is determined by the finite Toeplitz Gram matrix G_{j+1} .

2. Put $\mathfrak{D} = \mathcal{K}_+ \ominus V\mathcal{K}_+$ and

$$\widehat{\mathcal{K}} = \mathcal{K}_+ \oplus \mathfrak{D}_{-1} \oplus \mathfrak{D}_{-2} \oplus \cdots, \quad \mathfrak{D}_{-r} \simeq \mathfrak{D}.$$

If $\iota_ : \mathfrak{D} \hookrightarrow \mathcal{K}_+$ is the inclusion, then*

$$\widehat{U}(x, d_{-1}, d_{-2}, \dots) = (Vx + \iota_*d_{-1}, d_{-2}, d_{-3}, \dots) \quad (11)$$

is a minimal unitary dilation of the same moments.

Both constructions are valid without any nonsingularity or closed-range condition on a defect or Gram operator.

Proof. Positivity of the Gram forms, the isometric stage maps, their compatibility, and the moment identities were proved above. Equation (4) also gives

$$\mathcal{K}_+ = \overline{\text{span}}\{V^n e_0 h : n \geq 0, h \in \mathcal{H}\},$$

so V is minimal. The shell decomposition and $V\mathcal{K}_j \subseteq \mathcal{K}_{j+1}$ prove the asserted block form. All objects used for column j lie in E_{j+1} , proving finite determination.

Since V is an isometry,

$$\mathcal{K}_+ = V\mathcal{K}_+ \oplus \mathfrak{D}.$$

Consequently the two summands in the first coordinate on the right of (11) are orthogonal. The formula therefore preserves norms. It is onto as well: given $(y, e_{-1}, e_{-2}, \dots)$, decompose uniquely $y = Vx + \iota_* d$ and take the preimage $(x, d, e_{-1}, e_{-2}, \dots)$. Thus $\widehat{U}$ is unitary. For $h \in \mathcal{H} \subseteq \mathcal{K}_+$ and $n \geq 0$, its negative-tail coordinates remain zero, so $\widehat{U}^n h = V^n h$. Equation (4) proves the dilation identity.

It remains only to check two-sided minimality. The forward iterates of $\mathcal{H}$ span $\mathcal{K}_+$. Also $\mathfrak{D} \subseteq \mathcal{K}_+$, and for $d \in \mathfrak{D}$,

$$\widehat{U}^{-1}(\iota_* d, 0, 0, \dots) = (0, d, 0, \dots).$$

Approximating $\iota_* d$ by forward orbit vectors shows that the first negative copy of $\mathfrak{D}$ lies in the closed two-sided orbit of $\mathcal{H}$. Repeated application of $\widehat{U}^{-1}$ gives every further negative copy. Hence the closed span of $\{\widehat{U}^n h : n \in \mathbb{Z}, h \in \mathcal{H}\}$ is all of $\widehat{\mathcal{K}}$. $\square$

For completeness, the unitary matrix can be read directly from (11). Order the spaces as

$$\cdots \oplus \mathfrak{D}_{-3} \oplus \mathfrak{D}_{-2} \oplus \mathfrak{D}_{-1} \mid \mathcal{D}_0 \oplus \mathcal{D}_1 \oplus \cdots.$$

There are identity blocks from $\mathfrak{D}_{-r}$ to $\mathfrak{D}_{-(r-1)}$ for $r \geq 2$; the column from $\mathfrak{D}_{-1}$ into the positive side is

$$C_i = Q_i \iota_* : \mathfrak{D} \rightarrow \mathcal{D}_i \quad (i \geq 0);$$

and the positive-positive corner is exactly the Hessenberg matrix (8). All other negative-to-positive and positive-to-negative blocks vanish. This is the requested unitary block decomposition.

5 Verification, scope, and novelty

Degeneracy audit. The quotient by the radical makes every E_n well defined when G_n has a kernel. Completing the quotient handles nonclosed $\text{ran } G_n^{1/2}$. Isometric images of the complete stages are closed, so every shell and every orthogonal projection used above exists. If a shell or the wandering defect is zero, the same formula simply omits that summand.

Exact scope. The theorem answers the source question in canonical Gram coordinates for every $T \in \mathcal{C}_A$. It does not claim that the blocks can always be written as bounded rational expressions on fixed copies of $\mathcal{H}$; such expressions are precisely what can fail when the source recurrence asks for an inverse of a singular defect. Instead it gives an intrinsic finite-stage algorithm and a fully explicit action formula.

Bounded novelty check. On 11 August 2026 we searched the run’s registry and problem indexes for arXiv:2302.13873 and the core phrases “ $\mathcal{C}_A$ block dilation”, “operator-valued Toeplitz moment block Hessenberg”, and the exact wording of Question 5.9. We also performed bounded title/citation and web searches for later papers referring to the question. No later explicit answer was found. The positive-kernel/Kolmogorov construction and the minimal unitary extension of an isometry are classical, and the source itself discusses the abstract kernel route. The potentially new observation is the inverse-free finite Gram-shell extraction that turns this route into the requested general block model. Novelty confidence is therefore moderate, while proof-validity confidence is high.

Human-review recommendation. Check (i) the compatibility of J_n and S_n on quotient-completions, (ii) whether the canonical Gram-shell algorithm matches the intended meaning of “write block decompositions” in Question 5.9, and (iii) the minimality of the defect-tail unitary. No computational evidence is used in the proof.

References

- [1] B. V. Rajarama Bhat, A. Ghatak, and S. K. Pamula, *Operator moment dilations as block operators*, arXiv:2302.13873 (2023), Question 5.9.